📘 **AI/ML Internship – Day 13 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: What is Debugging?**

🔹 **Debugging Meaning**

Debugging is the process of:
👉 Finding and fixing errors (bugs) in code

🔹 **Types of Errors**

1. Syntax Errors

- Wrong Python syntax

```
print("Hello"
```

---

2. Runtime Errors

- Error while running program

```
x = 10 / 0
```

---

3. Logical Errors

- Code runs but gives wrong output

```
# wrong average
avg = total / (len(students) - 1)
```

📌 **Part 2: Debugging Techniques**

🔹 **1. Print Debugging**

```
x = 5
print("Value of x:", x)
```

🔹 **2. Step-by-Step Testing**

- Test small parts of code
- Check output at each step

🔹 **3. Input Testing**

- Test with different inputs:
  - Normal
  - Edge cases
  - Invalid

🔹 **4. Read Error Messages**

👉 Example:

ZeroDivisionError: division by zero

👉 Understand:

- Error type
- Line number

📌 **Part 3: Code Optimization (Very Important)**

🔹 **What is Optimization?**

Improving code to make it:

- Faster
- Cleaner
- Efficient

🔹 **Example: Normal vs Optimized**

❌ **Normal**

```
total = 0
for i in nums:
    total += i
```

✅ **Optimized**

```
total = sum(nums)
```

🔹 **Use Built-in Functions**

- sum()
- max()
- min()
- sorted()

📌 **Part 4: Writing Clean Code**

🔹 **Rules**

1. Use meaningful variable names
2. Keep code simple
3. Use functions
4. Avoid repetition
5. Proper indentation

🔹 **Example**

```
# Bad
x = 0
for i in l:
    x += i

# Good
total_sum = sum(numbers)
```

📌 **Part 5: Common Mistakes (Important)**

- Wrong indentation
- Forgetting type conversion

- Incorrect loop conditions
- Not handling exceptions

📌 Part 6: Real-World Debugging Example

```
def calculate_average(marks):
    try:
        return sum(marks) / len(marks)
    except ZeroDivisionError:
        return 0
```

📌 Part 7: Code Refactoring

🔹 What is Refactoring?

Improving code without changing output.

🔹 Example

Before

```
total = 0
for i in nums:
    total += i
print(total)
```

After

```
def get_sum(nums):
    return sum(nums)

print(get_sum(nums))
```

📌 Part 8: Real-World Thinking

👉 In AI:

- Bugs = wrong predictions
- Optimization = faster processing
- Clean code = maintainable models

🎯 **Tasks for Day 13**

🧠 **Task 1: Theory**

1. What is debugging?
2. Types of errors
3. What is optimization?
4. What is clean code?
5. What is refactoring?

💻 **Task 2: Debugging Practice**

👉 Fix errors:

```
# Error 1
print("Hello

# Error 2
x = int("abc")

# Error 3
nums = [1, 2, 3]
print(nums[5])
```

💻 **Task 3: Optimization Practice**

1. Replace loops with built-in functions
2. Optimize sorting
3. Reduce repeated code

💻 **Task 4: Refactoring**

👉 Convert:

- Long code → functions

- Repeated logic → reusable function